DESIGN SPECIFICATION / REVISION 6

Performance and trace harness

Measurement methodology, workload format, run modes and acceptance rules for svg-dom-graph

> **DECISION IN ONE SENTENCE**
>
> Build one harness with three measurement layers (pure Rust, WASM/DOM, real browser) and four run modes (correctness, performance, recovery, endurance), driven by a versioned workload manifest. It records raw observations and never lets a timing count unless the same run's checkpoint matches the reference model.

Prepared: 5 November 2026. **Target:** main at `a49c0bb`, crate 0.2.47. **Origin:** audit deliverable 1 / P0, and a review of revision 1 that found the timing, statistical, workload-format and layering rules missing. **Status:** Phases 0, 1, 2, 3, 4 and 5 complete (findings in `perf/README.md`, raw evidence in `perf/evidence/`, schemas in `perf/schemas/`). Revision 6 records the comparison, the baselines and the CI profile of section 5.5. Revision 5 recorded the Phase 4 contracts, the recovery runner and the endurance runners of section 5.4, and what breaking the library on purpose found. Phase 1 is described in section 5.1, Phase 2 in section 5.2, Phase 3 in section 5.3 and Phase 4 in section 5.4 and section 7.1, and Phase 5 in section 5.5.

> **WHAT THIS HARNESS CAN AND CANNOT CLAIM**
>
> It establishes a documented, repeatable operating envelope for named browsers, devices and workloads. It does not by itself prove production readiness, and a flat memory trend does not prove that every resource was released. A rising trend is a prompt to investigate, not a verdict of a leak.

1 Scope

- In scope: the measurement rules, the manifest and result schemas, the checkpoint contract, the runners, the statistical comparison and the CI profiles.
- Out of scope: the stress demos themselves, any new library API, line coverage, and Firefox or Safari timing in the first release.
- Kept outside the published library's runtime API. An instrumentation hook is added to the library only if a Phase 0 finding shows it is the sole way to observe something, and then as its own change.
- Reference workloads for validating the harness, before any large simulation: **W1** scene construction, **W2** connector rerouting during node movement, **W3** repeated scene creation and destruction (build on a fresh `<svg>`, drop the last `Scene` handle, remove the element).

2 Three measurement layers

A slow drag has several possible causes: connector maths, WASM-to-JavaScript calls, SVG attribute writes, browser layout and paint, or accessibility relationship updates. One latency number cannot separate them, so every workload is measured per layer, and results are never merged across layers.

Layer	What it times	Runner	Tells you where to optimise
A Rust	DOM-free work: connector geometry, label placement, layout, formatting, the demos' own algorithms.	Native binary in `perf-harness`, same statistics code as the other layers.	The library or application algorithm.
B WASM/DOM	Public `Scene` calls and their synchronous SVG mutations. Returns when the call returns. Not a frame time.	A harness wasm module loaded in a page; timed in page with `performance.now()`. Chrome through CDP first; Firefox later through the existing WebDriver setup.	svg-dom-graph, or svg-dom beneath it.
C Browser	Input to visual update, frame behaviour, accessibility tree, memory and listener counts.	CDP runner against the built gallery and the fixture, using real `Input.dispatchMouseEvent` input.	The browser, the page, or the application.

2.1 Layers are never subtracted

Layer A runs native machine code and Layer B runs WebAssembly, so the same algorithm costs differently in each. The difference of an A and a B figure is not the cost of the DOM or of the boundary, and no report presents it as one. A comparison may be shown as exploratory. Isolating DOM overhead would need matched workloads measured in one execution environment, and is not part of this design.

A Layer B figure is called a *synchronous API time* in every report. The words "frame", "render" and "paint" are reserved for Layer C.

3 Measurement boundaries

Each metric has one fixed start and stop. A runner may not report a metric under any other definition.

Metric	Starts	Stops	Layer and notes
Construction	Immediately before the first scene API call	The last synchronous scene API call returns	B. Excludes wasm instantiation, which is a separate cold-start metric.
Mutation	Immediately before the targeted API call (for example `Scene::move_node`)	The call returns	B. Includes Rust work, boundary crossings, SVG writes and connector updates, which cannot be separated at this layer.
Rust-only	Before the DOM-free function	It returns	A. Reported on its own. Never subtracted from a B figure (see 2.1).
Presentation	A user-timing mark written immediately before the mutation, in the browser's own clock	The frame that the correlation rule in 3.2 attributes to that mutation	C, Chrome only. Where correlation fails, the sample is `unavailable`. Elsewhere the fallback is the second `requestAnimationFrame` callback followed by a `MessageChannel` task, reported as *presentation proxy*, which shows a frame began and not that pixels were painted.
Interaction	The first call of a capture-phase listener for the input event, stamped with `performance.now()` in the page	The presentation boundary of the same sample	C. Both ends are in the browser's clock. The CDP command's round-trip time is recorded as a separate metric and is never used as the start. Event Timing entries are kept alongside, noting their 8 ms rounding.
Teardown	Immediately before releasing the scene	Synchronous cleanup returns	B. Resource counts are read after a settle step (section 6), not inside this interval.
Recovery	Immediately before fault injection	The documented consistent state is reached and its checkpoint matches	Recovery mode only. Reported as its own metric, never mixed into latency series.
Cold start	Navigation begins	Gallery ready	C. Separate from steady state.

3.2 Correlating a mutation with a frame

The next `Commit` or `Paint` event after a mutation is not necessarily the one it caused, because unrelated browser work can produce an earlier event. The runner therefore does the following. Phase 3's spike (`perf/spike/frame-correlation.mjs`) tested this rule against the page's own stamps of the frame callbacks that carried its mutations, and the rule below is the one that survived it. It differs from revision 3's in the four ways marked.

1. Records a CDP trace with the `blink.user_timing`, `devtools.timeline`, `cc` and `viz` categories, so the harness's marks and the frame lifecycle events share one clock. A mark's arguments carry the page's own `performance.now()` at the moment it was written, so the trace and the page are tied by the mark itself and no offset is assumed.
2. Writes a uniquely numbered mark immediately before each mutation, or in the first capture-phase listener for an input event.
3. Takes the main-thread frames to be the `ProxyMain::BeginMainFrame` events of the mark's process, each with the compositor's begin-frame id. **(Changed.)** The frame that carries the mutation is the one that *contains* the mark, if its style or layout step comes after the mark, since the browser delivers continuous input inside a frame (the spike saw it 40 times in 40). Otherwise it is the first frame that begins after the mark and has a style or layout step. A frame that has done its style step before the mark cannot carry it.

4. **(Changed.)** The presentation boundary is the end of the compositor's `PipelineReporter` whose `frame_sequence` is that frame's begin-frame id, that ended as `STATE_PRESENTED_ALL`, and that ended after the main thread finished the frame. With unrelated frames being made, Chrome also reports a `STATE_PRESENTED_PARTIAL` pipeline for the same frame, presented while the main thread was still at work, and taking it gave a presentation before the callback had run. A frame that was dropped is not presented, and is `unavailable`.
5. **(Added.)** A change that is followed by a forced layout, such as a panel that is built from HTML and then measured, has had its style and layout done in its own task, and the frame that shows it has only a paint. For changes that span tasks the qualifier is widened to a style or layout step *or* a repaint (`qualify: 'update'`). A frame that neither restyles nor repaints never qualifies. The wider qualifier can take an unrelated repaint for the one it seeks, so the gallery adapter uses it and the interaction runner does not.
6. Marks the sample `unavailable`, with its reason, when no such frame is found within a bound, when the mark falls after the style step of its frame and a later frame also has one, when the frame was dropped or only partly presented, when the mark is missing or written twice, or when the trace lost data.

What the figure is. A time from this rule is a *trace-attributed presentation latency*. Attribution is not causation: finding the frame whose lifecycle holds the relevant work does not prove that every change in it was caused by the marked operation, and under the wider qualifier an unrelated repaint can be taken for the one sought, so such a time is the upper bound of a window. And a compositor's presentation is not a photon: the display pipeline and scan-out are not in the trace, and headless Chrome presents without a display. Every Layer C metric states this in its mechanism, and no report presents these times as a guaranteed end-to-end latency.

Result of the spike. On a page whose callbacks have known costs, in Chrome 157: one mutation at a time, every sample was given the frame that holds its own callback, and none was presented sooner than its callback's cost; with unrelated work making a frame every 5 ms, every sample was still given the right frame; with a mutation every 6 ms and earlier frames still in flight, every sample was right, while the naive rule (the next presentation after the mark) was right for only about two thirds. A mutation that changed nothing was never given a frame. A trace whose buffer was too small reported data loss and gave no wrong frame. A mutation with its layout forced was missed by the style rule and placed correctly by the wider one. Real input from `Input.dispatchMouseEvent`, written at once or deferred to a frame, was placed in the right frame every time.

3.1 Timer resolution and batching

- `performance.now()` is coarsened by browsers. Phase 0 measured 100 µs in headless Chrome 157 and 5 µs when the page is served with COOP and COEP headers. The harness serves its own pages with those headers and records the measured resolution in every result.
- When an operation takes less than 50 times the measured resolution, it is timed in a **batch** of *k* operations and reported as batch throughput, with *k* recorded. A batch figure is never labelled as individual latency.
- Timing code runs outside the measured interval. The harness does not call `console`, allocate result objects or touch the DOM inside it.

4 Run modes

Modes are separate executions. Fault injection never runs in a performance mode, so a latency figure never contains an injected error or its recovery.

Mode	Purpose	Rules
Correctness	Verify outcomes and invariants against the reference model.	Checkpoint after every batch. Dangling endpoints, duplicate ids and model/DOM mismatch fail the run. No timing is kept.
Performance	Measure successful operations.	No faults. Timed regions contain no checks. The checkpoint is taken after the timed region and compared with the reference; a mismatch marks every figure from that run `invalid_incorrect` and excludes it from comparison.
Recovery	Exercise deliberate failures and verify eventual consistency.	Each fault point declares its documented commit contract (section 7). Recovery time is its own metric.
Endurance	Repeated construct / use / destroy cycles.	Resource counts and sampled memory over many cycles; trend reported with its mechanism (section 6).

5 Workload manifest and determinism

A manifest is a versioned TOML file. The same manifest must produce the same logical operation stream on every run, regardless of timing, machine or browser.

```
schema_version = 1
workload       = "connector_rerouting"     # W2
seed           = 12345
prng           = "splitmix64"              # fixed algorithm, implemented in the harness

[graph]
nodes = 1000
edges = 3000
maximum_degree = 16

[execution]
modes                  = ["correctness", "performance"]
warmup_iterations      = 100
measurement_iterations = 1000
repetitions            = 5               # independent runs, each a fresh browser tab

[measurement]
capture_dom_counts   = true
capture_heap         = false
capture_frame_timing = true
checkpoint_every     = 100               # correctness mode only

[limits]
dom_nodes = 50000
wall_ms   = 120000
```

Values here are illustrative, not recommended thresholds.

- **Own PRNG.** The generator is a small fixed algorithm in the harness, not an external crate, so a dependency upgrade cannot change a seed's meaning.
- **Operation-stream hash.** Each operation is encoded canonically and hashed with SHA-256 in order. The encoding begins with a version byte. Every value carries a one-byte type tag. A string is a little-endian `u32` byte length followed by UTF-8. An integer is little-endian at a fixed width. A float is its IEEE-754 bit pattern, little-endian, with `-0.0` written as `+0.0`. NaN and infinity are rejected when a manifest is read, as the library

rejects them. The hash is stored in the manifest's result, and a run fails if the executed stream hashes differently from the generated one. That proves the benchmark exercised the intended workload.

- **Reference model.** The generator also produces the expected checkpoint sequence from a DOM-free model, so correctness is checked against an independent computation.
- **Limits.** Exceeding one stops the run with `refused` or `timed_out`. Graceful refusal is never reported as a pass.

5.1 The workload engine (Phase 1)

The engine is the `perf-harness` crate. It reads a manifest, writes the operation stream, and drives a DOM-free reference model through it. It is described in full, with its evidence, in `perf/README.md`. This section records what the design now commits to.

Operation	Maps to
`create_scene`, `drop_scene`, `remove_host`	`Scene::new` on a fresh `<svg>`; dropping the last handle, which leaves the DOM; removing the `<svg>`.
`add_node`	`Scene::add_node`, a plain node at a size the caller gives, so the model knows the box exactly.
`add_edge`	`Scene::add_edge_with` with a straight connector. The library's default is an elbow, so a runner must ask for a straight one.
`move_node`	`Scene::move_node`.
`phase`, `checkpoint`	Markers. They change no scene, and a performance run skips the checkpoints.

- **Two digests.** A report carries `sha256` over every operation, which a correctness run executes, and `mutations_sha256` over the operations that change the scene, which a performance run executes. A result states which one it ran (`operation_stream_scope`). Neither depends on the repetition count, the limits or the capture options, and the mutations digest does not depend on where the checkpoints fall.
- **W1 and W3 differ in the graph.** W1 builds the same graph on every cycle and measures construction. W3 builds a different one each cycle, so that freed capacity has to be reused. W2 builds once and then moves nodes, one move in four aimed at the node of highest degree.
- **What the model predicts.** Topology, content, interaction, node boxes, connector ends and paint order are all predicted, and Phase 2 confirmed them against Chrome 157 for plain nodes and straight connectors. A straight connector's ends are where the ray between the two centres crosses each boundary, and where the two ends are one point the library draws no path and the model records none. Paint order is creation order, and a move does not raise a node. Ids and the accessibility tree are not predicted and are not compared.
- **Where it departs from section 5.** Coordinates are integers, so the float rule of the encoding is implemented and tested but no operation uses it yet. A seed written in TOML stops at $2^{63} - 1$.
- **Determinism.** The same manifest gives the same stream on native code and on 32-bit wasm, and an independent JavaScript generator gives the same counts and digests for 30 manifests. It has not yet been observed on a second physical machine, which the first run of the CI job does.

5.2 The runners (Phase 2)

Layer A is native and Layer B is a wasm module driving a real scene in Chrome. Both are described in full, with their evidence, in `perf/README.md`. This section records what the design now commits to.

- **Layers are reported apart.** Layer A times the library's own `boundary_point` for each connector at the moved node. Layer B times synchronous `Scene` calls. Neither is subtracted from the other, and neither is a

frame time.

- **Where the clock is read.** A timed interval holds library calls and nothing else. Each operation is prepared first (node ids, strings, the host element), the reference model is updated and the stream is hashed. Prepared operations of one kind wait in a queue that was allocated at full size, and the queue runs as one loop of `Scene` calls with `performance.now()` read once before the first call and once after the last. What the calls returned is kept after the second reading, and checks and resource counts run between queues. Moves are timed in batches sized from the warm-up so that one observation spans 50 timer steps. One clock read is inside each interval, and the outcome says what it costs. A timing under 50 steps that could not be batched is flagged `resolution_limited`.
- **What is observed.** The runner reads node boxes, edges, connector ends and paint order back out of the document, and the view from the scene, so a checkpoint is compared with what exists and not with what the runner meant. The comparison names the element, the coordinate, both values and the distance.
- **A performance run vouches for the end of the measured work.** Its two executed checkpoints are a completed scene and the empty page for W1 and W3, and the state after the last move for W2. A self-test applies the last mutation before each wrongly, and the run must come out invalid through a checkpoint.
- **A performance run still proves it was right.** It skips the checkpoints along the way and executes the last two outside its timing. Its digest is the mutations digest, and a run that did not execute exactly that stream is invalid.
- **The timing is itself checked.** A self-test delays the harness's own work by 100 to 200 us for every operation and fails if any timed figure moves. With a negative control: an earlier version of the executor timed the model update and the hash, and the same test showed it, reporting 207.5 us a move where the fixed one reports 6.1 us.
- **The checkpoint check is itself checked.** A self-test corrupts what was observed in six ways. Each must make the run invalid and name the right section, and a clean run must pass.
- **The counting allocator is verified before it is used.** It must add exactly the size of each retained block, return exactly on release, and count closures leaked with `Closure::forget`, including the library's own. The orchestrator checks it once per run and withholds every allocator figure if it fails.
- **The harness's own memory must not be measured as the library's.** Everything a run records is allocated at full size before the first reading, and a cycle that ran a checkpoint is left out of the endurance series. Without that, the harness's vectors doubling looked like a leak.
- **The endurance series carries its own cycles.** Each sample records its cycle number and whether that cycle was measured, and the end of the warm-up is taken from those and not from a position in a list that has had cycles removed. The net change of live bytes (how far it moved) and the least-squares slope against the cycle (how fast it moves) are separate metrics, the slope needs three points, and each metric states which cycles it covers.
- **What stays allocated after a teardown is accounted for.** A staged experiment, each stage in a fresh page, shows that the page, the library and wasm-bindgen's glue leave nothing behind for a scene of up to 24 nodes. A tracking mode of the allocator, which follows every block allocated after the baseline by the phase of the harness that made it, found the rest in a real endurance run: one block that doubles with the peak number of JS handles (512 bytes at 40 nodes, 1,536 at 80, 3,584 at 160, 7,680 at 320), the hash's 64-byte encoder buffer, and the generator's own plan for the next cycle, which was removed. The one-time cost is reported apart from the endurance interval.
- **The endurance gate is itself checked.** It fails when neither the net change nor the carried slope of live bytes is within `limits.live_growth_allowance_bytes` (zero if absent), and too few cycles to judge is inconclusive. A self-test puts leaks of known size into every cycle: plain bytes, a leaked closure, and the library's own `svg_buttons::watch`. Every one fails the gate, the slope equals the size of the leak, a clean run passes, and the allowance turns the verdict at exactly the growth that was seen.

- **Limits.** The vocabulary registers no listeners, so the listener gate holds at zero and proves little for W1 to W3. The interaction workload of section 5.3 makes nodes draggable, and its listener gate counts 105 listeners for 24 nodes. Firefox was not run.

5.3 The browser runner (Phase 3)

Layer C drives a real scene in Chrome with real input and reads what the browser did. Both parts are described in full, with their evidence, in `perf/README.md`. This section records what the design now commits to.

- **A fourth workload.** W4, `interaction`, builds one scene, checks it against the reference model, and leaves it for a browser. Its manifest has an `[interaction]` table: the scales of the view in percent (a manifest holds no floating-point number), the gestures (drag a node, pan the background, wheel-zoom with the control key held), the pointer moves in a drag and how far each goes. The same seed gives the same plan on native code and on 32-bit wasm, and the plan of the reference manifest has a fixed digest. The independent JavaScript generator agrees on the stream.
- **Every gesture says what it must do.** The plan states, for each gesture, where the pointer goes down and what the scene must look like afterwards: a node moves by the pointer's movement over the scale, a pan translates the view by the movement and does not scale it, and a wheel notch multiplies the scale by 1.25 about the pointer. After every gesture the scene is read back and compared. Chrome 157 agreed with every prediction at 50%, 100% and 200%. A gesture that did not do what it was meant to makes the run invalid, since a time for a gesture that did nothing is not a measurement.
- **Where the interaction time starts and stops.** It starts at the first capture-phase listener for the input event, stamped with `performance.now()` in the page, which also writes the mark. It stops at the presentation boundary of the frame that section 3.2 attributes. The CDP command's round trip is a metric of its own and is never a start. The browser's input delay (the event's `timeStamp` to the listener) and its Event Timing entries are kept beside it.
- **The wheel is on its own.** A real wheel reaches only the tab in front, so each wheel case runs in a tab of its own that has been brought to the front.
- **Draggable nodes make the listener gate mean something.** The scene's nodes are draggable, so a built scene holds listeners (105 for 24 nodes), and the gate that they return to the baseline after the teardown now fails if they do not. A gate that passes because nothing was ever counted is reported as inconclusive.
- **Chrome's accessibility tree is checked.** In a tab of its own, after the timed cases and never inside a timing, the runner reads `Accessibility.getFullAXTree` before and after a drag, a pan and a wheel zoom. A node with an edge must be a group named for its key with the model's `Output to` and `Input from` clauses, the graph view must be one application whose name carries the zoom, Tab must reach the graph view and the tree and the DOM must agree on which element has focus, names must not change when a node moves, and a torn-down scene must leave nothing in the tree. At most 40 nodes are inspected, and the tree's size is bounded. The 56 checks of the small manifest and the 276 of the medium one pass, and a self-test that strips one node of its relationships makes the run invalid.
- **The harness is itself checked.** A self-test adds a listener that costs 20 ms after every move, and the interaction time goes up to match while the run stays valid. It stops the input before the library sees it, and the run comes out invalid. It makes every CDP command 30 ms slower, and the round trip goes up by that while the interaction time does not move.
- **The gallery adapter** selects every panel with a real click on its menu entry, so it works for every panel without a change to any of them. It reports the first visit and a revisit, from the click to the end of the frame that shows the panel (by the wider qualifier), the DOM and listener counts either side, the page's errors, and a walk of the panels that have Next, Previous or Restart controls, found by accessible name. A panel with a manual clock is advanced through its hook, named in `perf/gallery-overrides.json`. A missing control downgrades the panel and does not fail the sweep. **Each walk gets a verdict of three outcomes.** Pass: the control exists, its expected effect was seen and the invariants defined for the panel hold. Inconclusive: a

control is absent, or no expectation is defined, or the panel declares that it has none. Fail: a defined expectation was broken, and the sweep exits 1. What every panel is held to is small (an enabled Next or Previous changes the panel, an enabled Restart changes a moved panel, the page throws nothing). Whether Previous retraces Next from each step, and whether Restart returns to the first state, are held only where a panel declares them with the place in its source that says so, and an observed mismatch of anything undeclared is written down and never counted as a pass. In the recorded sweep 16 walks passed, 38 were inconclusive and none failed, and a self-test shows that a wrongly declared expectation fails. The gallery builds each panel once and only hides it afterwards, so none of this is a teardown, and the report says so. Its report has a schema of its own.

- **Limits.** Chrome only, on one machine, headless, and the compositor presents without a real display. The input cadence (one move every 24 ms) is a choice of the harness. A frame that the compositor dropped is `unavailable` and not guessed at, which loses most of the steps of the heaviest panels. Accessibility is checked in Chrome's tree for a bounded sample, with no screen reader. The gesture plan is not cross-checked by an independent generator, though its digest is the same on native code and on 32-bit wasm. The pointer is moved to the nearest place that reaches the node or the pan surface when a connector is drawn over the planned one, and the sample says how often. Firefox was not run.

5.4 Recovery and endurance (Phase 4)

Both are described in full, with their evidence, in `perf/README.md`. This section records what the design now commits to.

- **A fifth workload.** W5, `recovery`, builds one scene and checks it, as W4 does, and leaves it to be broken. Its manifest has a `[recovery]` table of fault points and their expected outcomes.
- **A failure is put at every place an operation can have one.** An injector in the page makes the Nth DOM call that the library makes and that can fail throw. An operation is run once on a twin with no fault, to learn how many calls it makes and what it leaves, and then once for each of those calls with that call failing. What is left is compared, as a checkpoint and as the document itself (element counts, listener counts, paths, label positions, styles), with the state before and with the reference, which the reference model must agree with. A retry is made as the contract defines it, and must reach the reference. A call that cannot report a failure in the bindings is not injected.
- **Recovery time** is its own metric, taken in the page around the failed call and the call that recovers from it, with no check inside the interval. It is never mixed into a latency series.
- **Layer C breaks what needs a frame or input:** a refused frame and a failed write in a frame, a scene dropped with a frame pending, and a scene dropped between a pointer down and a pointer up, with real input.
- **Endurance under failures.** Each cycle breaks each fault point at one DOM call (a different one each cycle), recovers, takes the scene down and reads the document's elements and listeners, the counting allocator's live bytes, and in Layer C the heap after a forced collection. Elements and listeners must return to their baseline after every cycle, every recovery must reach its promised state, and live bytes must not grow after the warm-up.
- **What it found.** Every documented contract held at every DOM call (45 failures in Layer B, 40 scenario runs in Layer C). Two findings were defects and were fixed. A node whose scene was dropped in the middle of a drag kept its grabbing cursor, and now has it taken off, which the Layer C scenario judges. The other: `add_edge_with` had no rollback after its connector was attached, so a failure left a stray connector, or one end named and not the other, and a retry added a second connector. It failed at 6 of its 10 DOM calls. The library now makes it a transaction, the same sweep against the original library fails those 6 and against the fixed one none, a test in the library's own browser suite makes every one of its writes fail in turn, and the point moved from undocumented to `nothing_committed`. Two things remain that nothing documents. A removed host is not an error, and the library now says that a detached scene is unsupported. And a failed

DOM call that returns an object costs wasm-bindgen one externref slot for good, exactly one each, which the endurance run found as a step in live bytes at cycle 141 and which the runner now counts and explains exactly instead of tolerating.

- **The runner is itself checked.** The injector fires only where it is armed. Each documented point, given the other outcome, fails. A library with no rollback and a frame that is not cancelled are both caught. A leak of bytes, a listener, an element and an unexplained lost slot each fail their own gate and no other.
- **Limits.** A failure is a thrown DOM call, in Chrome, on one machine. Label placement depends on history, so a connector retry is held to the connectors. Not covered: a label that cannot be drawn, the elbow connector, and the nested-scene, operator and data-node constructors. Hour-scale runs are the same manifest with more cycles.

5.5 CI integration, baselines and regression reporting (Phase 5)

Described in full, with its evidence, in `perf/README.md`. This section records what the design now commits to.

- **The comparison.** The unit is the repetition. A run gives one figure for a metric, a percentile of its own observations, and a collection gives the median of its runs' figures. The statistic is the ratio of the candidate's to the baseline's. Its 95% interval is a hierarchical bootstrap of 4,000 resamples from a fixed seed: repetitions are drawn with replacement from each side, then the observations of each chosen one. The same figures give the same interval to the last bit, on 64-bit native code and on 32-bit wasm.
- **Verdicts.** With a ratio tolerance of 10%: `regression` if the interval is wholly above 1.1, `improvement` if wholly below 1/1.1, `no_material_change` if wholly inside both, and otherwise `inconclusive`, which is never a pass. With fewer than 5 repetitions on a side, or too few observations for the percentile, the result is `diagnostic`: the medians and ratio are shown, with no interval and no verdict. Two sides that were not measured alike, or a run that was not valid, are `not_comparable` for every metric and compare no figure.
- **A baseline is written by one command.** `promote-baseline.mjs` copies the runs and writes `promotion.json` (commit, workload hash, environment key, method version, reason, what it replaced). A run, and a comparison, never touch a baseline. It refuses an invalid run, a run of a candidate changed on purpose, a tree with uncommitted changes, more than one commit, manifest or environment, too few runs, and a regression against the baseline it would replace unless a reason is given, which is recorded with the metrics.
- **perf-smoke** runs on every commit: W1 to W3 at a size that gives a median at least 30 observations, and W4 small, with 3 repetitions. It fails on the deterministic gates only, and never on a timing. The comparison with a baseline is written to the job summary and kept as an artifact for 90 days; with 3 repetitions it is diagnostic. Statistical verdicts block a change only from a baseline of 10 runs on one runner class, and `--enforce-statistical` is not given until there is one.
- **The exit condition is tested.** On synthetic runs with the real binary and the real command (per commit, no browser): a clean fixture is no material change, a slowdown of 35% is a regression, a speedup an improvement, every refusal refuses, and a comparison leaves a baseline as it was. In Chrome: five clean runs promoted, five more not a regression, five with a slowdown injected inside the timed interval a regression (ratio 7.8), and a clean lifecycle passes while the same with listeners leaked fails on 90 listeners left.
- **Calibration.** Over 100 trials per size, with a run-to-run spread of 3% and a within-run spread of 10%: no false alarm at no change, a change of 5% never a regression, one of 15% found 74% of the time, and one of 35% every time. The figures belong to the assumed spread.
- **What perf-smoke found.** A performance run that times nothing (Layer A for W1 and W3, Layer B for W4) failed the result schema. Such a run is now said and is not a result.
- **Limits.** Only p50 is compared by the smoke manifests. One tolerance serves every metric until a baseline of 10 runs gives an observed spread. Baseline and candidate runs are not alternated, since the baseline is a committed earlier set. No baseline is committed yet, because the one that matters is the CI runner's. Nothing was run on a CI runner.

6 Statistical procedure

- **Phases.** Cold start is measured once per repetition and reported separately. A warm-up phase of fixed iterations is discarded from steady-state statistics but its raw values are kept. The measured phase has a fixed iteration count, never a fixed "until stable" rule.
- **Repetitions.** Each repetition is a separate experimental run in its own Chrome process with its own profile directory. They share the machine, its load and its thermal state, so they are not fully independent environments, and reports say so. For a baseline comparison, runs of baseline and candidate alternate (ABAB) so that drift is shared. Variation between repetitions is reported separately from variation within one.
- **Raw data.** Every observation is written to a samples file next to the result. Nothing is dropped. Outliers are flagged (Tukey fences, 1.5 × IQR) and counted, never removed.
- **Percentiles.** p50 needs at least 30 samples, p95 at least 200, p99 at least 1000. Below that, the report gives only min, median and max and says why. A percentile is given with a 95% interval from a hierarchical bootstrap: resample the repetitions, then resample the observations within each chosen one. The sample count sits beside it. These minimums set when a figure is shown and do not guarantee a reliable interval.
- **Comparison.** The unit is the repetition. The statistic is the ratio of candidate to baseline repetition medians, with a bootstrap interval across repetitions. With a ratio tolerance δ the verdict is `regression` (interval wholly above 1 + δ), `improvement` (wholly below 1 / (1 + δ)), `no_material_change` (wholly inside both bounds) or `inconclusive`. Inconclusive is never reported as a pass.
- **Few repetitions.** With the three repetitions of `perf-smoke`, results are labelled diagnostic and no verdict is issued.
- **Same environment.** A comparison is made only between runs whose keys match: OS, CPU model, runner class, the *full* browser version, and the harness's measurement protocol version. Otherwise the report says "not comparable".

6.1 Resources and memory

Kind	How measured	Cross-browser?
Deterministic counts	SVG element count by kind through the DOM; a listener tally; outstanding callbacks and scene-owned objects where the harness page can observe them. The tally counts effective registrations: it keys each by target, type, listener and capture flag, ignores a duplicate as the platform does, and handles `once`, `AbortSignal` and removal. It is cross-checked against Chrome's own count. It is installed before the wasm module loads.	Yes. These are the gate-eligible figures. Scene-owned listeners are released when the last `Scene` handle drops, so the gate is the count after a drop equalling the count before the build (see `perf/README.md`, finding 7).
Wasm capacity	The byte length of the module's linear memory, as a `capacity` metric: a high-water mark. It grows in pages and never shrinks, so it shows what was ever needed and not what is live. Reported as a diagnostic, apart from heap and listener figures.	Yes, in Layers B and C.
Wasm live allocation	Live bytes and allocation count from a counting global allocator in `perf-harness-wasm`, which the library links into unchanged. The library deliberately leaks some closures (`Closure::forget`), and this is the figure in which that would show. Not available in the demo gallery's module.	Layer B only. A design to be proved in Phase 2.
Browser counters	Chrome `Performance.getMetrics` (`Nodes`, `JSEventListeners`, `JSHeapUsedSize`) after `HeapProfiler.collectGarbage`. Measurement mechanism recorded with every figure.	No. Chrome-only diagnostics. Never an acceptance criterion in other browsers.
Endurance trend	Counts and heap sampled at the end of each cycle after a settle step (forced GC and two frames). Trend is a least-squares slope with its interval.	Counts yes; heap Chrome only.

Phase 0 found no `on*` property handlers in the library. The strongest lifecycle evidence combines count assertions, the existing listener-teardown tests and repeated create / destroy.

6.2 Reading wasm memory

Linear-memory size is capacity, not live Rust memory. A constant size can conceal retained allocations, and growth does not establish a leak. A lifecycle test therefore does not expect the size to return to its starting value after teardown. After a warm-up it counts how often memory grew during the measured cycles, which should be zero once capacity is being reused, and it compares live bytes after teardown with live bytes before the build. Growth in capacity with flat live bytes is reported as an observation, never as a leak.

6.3 Teardown observation point

Forced garbage collection is a Chrome diagnostic. It is not required for correctness and does not guarantee that every object is reclaimed. In Phase 0, Chrome's listener count read 139 before a collection and 37 after. Counts are compared with a baseline captured at the equivalent lifecycle point, for example the same scene built once and not yet dropped. The library has no public teardown call, so a teardown here means dropping the last `Scene` handle and removing the host `<svg>`. Leaving a gallery panel is not a teardown, because panels are built once and only hidden.

7 Checkpoints and failure contracts

A checkpoint is a structured snapshot with separately comparable sections, defined in `perf/schemas/checkpoint.schema.json`. Full numeric values are kept. A mismatch report names the section, and for geometry gives the element, coordinate, expected value, actual value and absolute difference.

Section	Content	Comparison rule
Topology	Nodes and edges by logical key (the name the scene was given), with endpoints; duplicate and dangling findings.	Exact, with a hash for diagnosis. Compared by logical key, not by DOM id.
DOM identity	Set of rendered element ids; uniqueness.	Uniqueness is always checked. Equality across scenes is not, because Phase 0 showed the arrow marker id comes from a process-wide counter (`svg-dom-graph-arrow-N`).
Geometry	Node boxes, connector paths, label positions in scene units.	Compared directly within 0.01 units on the full values. Never hashed: rounding before hashing can split two values that are well within tolerance (10.0049 and 10.0051).
Content	Data values, labels and text as displayed.	Exact.
Interaction	Selection, focused element, view transform.	The manifest states one of three things: the view is expected unchanged, the view is expected to end at a stated transform (a deterministic pan workload verifies it), or it is ignored.
Accessibility	Roles, names, relationships (Chrome AX tree in Layer C; DOM attributes elsewhere).	Exact, compared separately from the rest.
Paint order	Logical keys in paint order, where z-order matters.	Order significant, compared by logical key and not by DOM id. Other child ordering is sorted away.

Hashes are stored for the exact sections only, to identify which one differs.

7.1 Recovery mode

A failure before a change commits is not the same as a failure after part of it has. For each fault point the manifest names one of two expected outcomes, taken from the library's documented behaviour:

- **Nothing committed:** the checkpoint equals the one taken before the operation.
- **Partly committed:** a documented set of sections differs (for example, the node moved but its connector did not follow), and a defined retry brings the checkpoint to the reference.

Fault points: node construction, DOM transform, connector update, label update, drag installation, edge construction, scheduled frame, teardown, mid-drag teardown and a removed host SVG. Phase 4 began by

tabulating each point's actual contract from the code, in `perf/fault-contracts.json`, with the lines that state it. Nine have a documented contract: node construction, DOM transform, drag installation and edge construction promise that nothing is committed, connector update and label update promise that the move is kept and that a defined retry puts the rest right, the scheduled frame promises that a refused or failed write is tried again by the next event, and teardown and mid-drag teardown promise that the pending frame is cancelled and the handlers do nothing. **One has none**, a scene whose host has been removed, and is not tested against a guess. Edge construction was the other, until the first sweep of it found that `add_edge_with` was not a transaction (section 5.4) and the library was fixed, so that it now has the contract that it was found to lack. A recovery manifest names each point with `nothing_committed`, `partly_committed` or `undocumented`, the third being how the rule of this section is written into a manifest, and a test checks that the manifests agree with the contracts and that every line the contracts cite exists.

8 CI profiles and acceptance

Profile	Content	When
`correctness`	Layer A and B small workloads; checkpoints; deterministic gates.	Per commit, blocking.
`perf-smoke`	W1 to W3 at a size that gives a median 30 observations, and W4 small, 3 repetitions, Layers A to C where they apply.	Per commit. Blocking on the deterministic gates; the baseline comparison is **reporting only** (section 5.5).
`gallery`	Every gallery panel loaded and revisited; steppable demos walked.	Nightly.
`endurance`	Hour-scale cycles with sampled counts and heap.	Weekly or manual, on a fixed runner.

8.1 Gates

- **Deterministic gates (blocking from the first release):** checkpoint correctness, DOM and listener counts at teardown returning to baseline, resource limits, malformed-manifest rejection, the operation-stream hash.
- **Statistical gates (reporting only until at least 10 baseline runs exist on one runner class):** section 6's verdicts. Thresholds are then set from the observed run-to-run interval, not guessed.

8.2 Results and history

One JSON result per run, and a separate comparison report per pair of run collections. A result holds no between-run statistic or interval, since one run has neither. It carries a complete copy of its manifest, so it stands alone, with the manifest hash and the operation-stream hash. A result has `schema_version`, environment (browser and version, OS, CPU, memory, rustc, wasm-pack, crate version, git SHA, dirty flag, timer resolution), manifest copy and hash, mode, status (`ok`, `invalid_incorrect`, `refused`, `timed_out`, `error`), the metrics of section 3 with sample counts and within-run spread, and a pointer to the raw samples file. Each metric has its own status, `measured`, `unavailable` or `measurement_error`, so a trace or heap sample that fails does not turn a correct workload into a failed one. The schemas are `perf/schemas/manifest.schema.json` and `result.schema.json`. They are strict: a fixed object rejects unknown keys, and a metric's status decides which fields it must or must not carry (a measured one needs its unit, mechanism, sample count and summary; an unavailable one needs a reason and no figures). Validation is a deterministic gate, and `perf/schemas/check.mjs` runs every valid example and every deliberately broken one, so a schema that becomes too permissive fails.

The comparison report (`comparison.schema.json`) names the baseline and candidate runs and commits, and records the baseline's promotion record, the compatibility key, the bootstrap settings, the ratio tolerance and the confidence level. For each metric it gives each side's repetition count, median and interval, between-run and

within-run spread, the ratio with its interval, and the verdict. Baselines are committed under `perf/baselines/` only by an explicit promotion command, which records the commit, the workload hash, the environment and the comparison method. A baseline is never updated by a run, and never after a regression without that command. Historical runs are CI artifacts, not committed, and a comparison report lists the baseline and candidate SHAs.

9 Existing demos

The gallery adapter lives in the Layer C runner. It selects a panel through its menu entry, so it works for every panel without changes. A panel with Next, Previous or Restart controls, found by accessible name, is also walked (forward then back, restart), and one with a test clock is advanced deterministically. A panel without them is measured for load and revisit only. Per-panel overrides are allowed where a control name is ambiguous, and a missing control downgrades a panel instead of failing the sweep.

Limit found in Phase 0. The gallery builds each panel once and later only hides it, so selecting and leaving a panel is not a teardown. The adapter reports build cost, revisit cost and the counts after a visit, and says so. Teardown is measured only by W3 and by the demos that stop their own clocks on `pagehide` (N-Queens, Tarjan, springs).

10 Phases

Phase	Deliverable	Status and exit condition
0	Methodology, schemas, spike.	**Complete.** Schemas are in `perf/schemas/` and the four spike answers are in `perf/README.md`: panels are not torn down on leaving; ids are not stable across scenes; there are no `on*` handlers; timer resolution is 100 µs, or 5 µs when isolated. It also found that leaked closures are invisible to the JS heap, and one listener result that a registration tracker then explained: a scene's listeners belong to its `Scene` handle and are released when it drops, while the DOM stays.
1	Deterministic workload engine and reference model for W1 to W3.	**Complete.** The `perf-harness` crate, the workload-report schema and three reference manifests. The same seed gives identical operations, digests and checkpoints on native code and on 32-bit wasm, and an independent JavaScript generator agrees. Not yet seen on a second physical machine (section 5.1).
2	WASM/DOM runner (Layer B), Layer A runner.	**Complete.** The `perf-harness-wasm` crate, the Layer A runner, the orchestrator and the result schema's native environment. Repeatable figures: over five fresh-Chrome repetitions the construction medians of a 200-node build agree to within 3% (nodes 3%, edges 2%, total 1%). The 1,000-move run gives 40 batched observations a repetition, whose medians span 6.7 to 7.7 us (14%), and have spread by 5% to 16% across recordings, so no p95 is reported. Timed intervals hold library calls only, which a timing self-test proves. Correctness-mode checkpoints pass at 1,000 nodes and 3,000 edges. The counting allocator is verified with a deliberately retained allocation, a release and a leaked closure (section 5.2).
3	CDP runner (Layer C) and gallery adapter. *Conditional on a spike that proves the frame correlation of 3.2.*	**Complete.** The spike held, with four changes to the rule (section 3.2). The interaction workload W4, the Layer C runner with drag, pan and wheel zoom at three scales (the wheel in a tab of its own), resource counts with a listener gate that now counts something, a Layer C self-test, and the gallery sweep, which loaded all 54 panels and walked 16 (section 5.3).
4	Recovery and endurance runners.	**Complete.** Each fault point recovers according to its documented contract, or is listed as a finding: 9 contracts held at every DOM call, two defects were found and fixed, and 2 findings are reported (section 5.4). The runner, its injector and its endurance cycles are checked by a self-test that makes each of them fail.
5	CI integration, baselines, comparison and history.	**Complete.** A deliberately injected regression and an injected listener leak are each caught by the harness's own tests, in a browser and on synthetic runs, and a clean candidate produces neither (section 5.5). The statistical gates report only until a baseline has 10 runs on one runner class.

10.1 Testing the harness

- Native unit tests for percentiles, bootstrap intervals, trend slope, Tukey flags, the PRNG, the stream hash, the schema and the comparison verdicts, run with the per-commit tests.
- Known-answer tests: a seed's first values and stream hash are fixed in the repository.
- Every test returns `Result<(), String>`, following the crate's convention.

11 Decisions

Question	Decision
Crate layout	Two on-demand workspace members: `perf-harness` (manifests, workload generation, statistics, reports) and `perf-harness-wasm` (Layer B, consuming the shared workload definitions and the public API). Manifests in `perf/manifests/`. Neither adds a dependency to the published library.
Baselines	`perf/baselines/`, written only by an explicit promotion command.
`perf-smoke` in CI	Done in Phase 5. It fails on an incorrect checkpoint, a leaked deterministic resource or a malformed workload. It never fails on a statistically uncertain timing comparison.
Chrome-only memory	Accepted as diagnostic evidence. Counts and wasm memory are the cross-browser figures.
TOML	Accepted, confined to the harness tooling.

11.1 Still open

- Whether the Phase 0 findings hold in Firefox and Safari. Nothing was run in either.

svg-dom-graph / performance harness / design specification rev 3. Derived from the 4 November 2026 coverage audit and two rounds of review. Thresholds are set from measurement, not from this document.